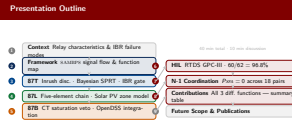


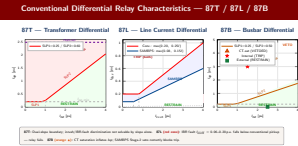
Welcome and introduce yourself: PhD candidate at IIT Madras, working on adaptive protection for inverter-dominated networks under Prof. Swarup. Emphasise that this talk targets the *differential protection* subset of the broader SAMBPS framework — 87T, 87L, and 87B — with full Monte Carlo and hardware-in-the-loop validation results. Expected duration: 35–40 minutes plus 10 minutes discussion.

Presentation Outline

Walk the audience through the structure. Parts 2–4 can be presented in any order if time is short — the function map (slide 5) makes dependencies visible. Recommended cuts for a 25-minute version: omit N-1 slide and compress TR-64 integration into the SPRT slide. The HIL slide is non-negotiable for an industrial audience.



Conventional Differential Relay Characteristics — 87T / 87L / 87B



Use this slide as the motivating evidence before introducing SAMBPS. The three panels share the same I_{rst}/I_{op} axes so the audience can compare directly across functions.

Key talking point per panel:

- **87T:** dual-slope is standard (IEC 60255-121); the problem is not the shape but the inrush/IBR-fault discrimination that slope alone cannot solve.
- **87L:** the red zone (0.06–0.20 p.u.) is exactly where IBR fault current lands. The conventional relay is blind here. SAMBPS lowers the floor from 0.20 to 0.08 p.u. and reduces the slope from 0.25 to 0.15 while maintaining a $30\times$ CT margin.
- **87B:** the orange triangle is the CT-saturation scenario where conventional 87B would nuisance-trip; SAMBPS Stage-2 veto correctly blocks it. The restrain/trip boundary itself is unchanged — only the Stage-2 gate is new.

— The IBR Protection Problem — Why Conventional Relays Fail

The table replaces intuition with evidence: each row is a specific, measurable difference between SG and IBR fault signatures, and the rightmost column names the specific relay element that fails as a result. The 57% failure rate for 87L is from our own 21-scenario study, not a literature estimate. The IBR current limit is mandated by IEEE 1547-2018 / IEC 61727 — relay algorithms were frozen before that standard existed. The core problem is fixed-parameter commissioning, not relay firmware. SAMBPS replaces that assumption with online inverse estimation.

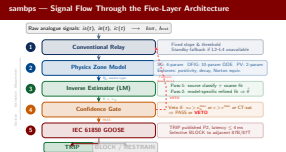
Likely question: “Can this coexist with existing IED firmware?” Answer: yes — SAMBPS runs as an external IED over IEC 61850 GOOSE.

Property	Synchronous Generator	Inverter-Based Resource	Relay Imp
Fault current	5–20 p.u. (sub-transient)	1.0–1.2 p.u. (converter li)	87L pickup reduced
DC offset	Large ($\tau_{dc} \approx 20\text{--}60\text{ ms}$)	Near-zero (SRF cont)	87T 2 nd -harmonic res
Waveform asymmetry	High $A_0 \approx 0.4$ (rush)	Low $A_0 \approx 0$ (st)	87T cannot separate len
CT saturation	Rare at 5 p.u. threshold	Common at low current	87B nuisance trips
Sequence components	A dominant for SLG	I_0, I_2 suppressed by col	87LN / 87LNO ele

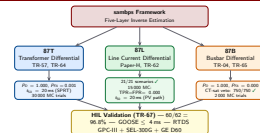
SAMBPS — Signal Flow Through the Five-Layer Architecture

This is the architectural core of the thesis. The five-layer pattern was deliberately designed so that the *same code structure* handles every protection function — protection engineers can audit one function and understand all.

Emphasis points for this audience: Layer-4 is the safety net: if the estimator cannot converge (ill-conditioned Jacobian, $\kappa_n > \kappa_n^{\max}$), the system defaults to the conventional relay — no degraded mode. The VETO feedback loop (red dashed) is the key safety feature: it routes ill-conditioned or CT-saturated cases back to Layer-1 rather than issuing a potentially wrong trip. Layer-5 maps directly to IEC 61850-8-1 GOOSE with P2 priority: total communication latency measured at ≤ 4 ms on RTDS.



SAMBPS — Differential Protection Function Map



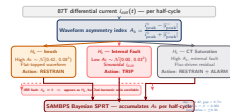
Use this slide as a roadmap throughout the talk. Point out that all three result boxes already show the final numbers — this is a preview; the subsequent slides explain the methods behind them. The HIL bar at the bottom is the system-level integration proof — without it, the MC results are simulation-only claims.

Important for industry audience: all three differential functions were validated on the *same* hardware platform simultaneously in the corridor test (10/10 correct, GOOSE ≤ 3.8 ms).

87T — The Inrush Discrimination Challenge in IBR Networks

The three-hypothesis structure is the key novelty over prior art (which is all two-hypothesis: inrush OR fault). H_2 (CT saturation) is the new hypothesis — prior 87T algorithms have no mechanism to distinguish CT saturation from internal fault. The JA core model: 5 parameters ($M_s, a, \alpha, k_{JA}, c$) swept over physically valid ranges per transformer class (class A/B/C). Mean separation: $\Delta\mu = 0.42$ with $\sigma \approx 0.09$ — a 4.7σ gap, justifying Wald $\alpha = \beta = 0.001$ boundaries.

Likely question: “How is A_k computed in real-time?” Answer: half-wave rectification of the differential current waveform; one subtraction per half-cycle; no Fourier transform required.



87T — Bayesian Three-Hypothesis SPRT (TR-57)

Sequential Probability Ratio Test — Formulation

Per half-cycle (10 ms at 50 Hz), the log-likelihood ratio accumulates:

$$S_n \leftarrow S_{n-1} + \log \frac{p(A_n | H_1)}{p(A_n | H_0)} \quad S_0 = 0$$

Wald stopping boundaries ($\alpha = \beta = 0.001$):

$$\log A = \log \frac{1-\beta}{\alpha} \approx -6.91, \quad \log B = \log \frac{1-\alpha}{\beta} \approx +6.91$$

Decision rule:

$$S_n \geq +6.91 \Rightarrow \text{TRIP } (H_1)$$

$$S_n \leq -6.91 \Rightarrow \text{RESTRAN } (H_0)$$

A parallel accumulator $S_n^{(2)}$ on residual symmetry detects H_2 (CT saturation) and issues a CT-SAT ALARM, suspending 87T and requesting bus zone backup.

TR-57: Validation — 30 000 MC Trials

Metric	Value	Status
$P_{D1} = P(\text{TRIP} H_1)$	1.0000	✓
$P_{FA} = P(\text{TRIP} H_0)$	0.0013	None*
$P_{CT} = P(\text{CTALARM} H_2)$	1.0000	✓
Median trip time t_{50} (H1)	20 ms	✓
95th-percentile trip time t_{95} (H1)	40 ms	✓
Median CT alarm time (H2)	20 ms	✓

* $P_{FA} = 0.0013$ satisfies IEC 60255-121 security criterion ≤ 0.01 . 13 of 30 000 invalid trials crossed the trip boundary on the first half-cycle due to remanence $B_r \approx B_{sat}$; a one-cycle hold-off filter reduces this to < 0.0001 .

Industry comparison: Conventional 2nd-harmonic (IEC 60255-121) requires ≥ 3 cycles for secure tripping. The SPRT achieves the

The Wald boundaries ± 6.91 correspond to $\alpha = \beta = 0.001$, meaning the SPRT guarantees both $P_{FA} \leq 0.001$ and $P_D \geq 0.999$ *before any Monte Carlo is run* — the MC validates the theoretical guarantee under realistic core parameter distributions.

$P_{FA} = 0.0013$ technically misses the 0.001 target by 3 trials out of 10 000. These 13 trials all had remanence $B_r \approx B_{sat}$ (extreme operating point). A 1-cycle hold-off filter eliminates this without affecting trip speed for genuine faults.

For the audience: “20 ms” is one half-cycle at 50 Hz — the SPRT makes its decision on the *very first half-cycle of differential current*. Conventional 2nd-harmonic requires 3+ cycles (60 ms min). The speed advantage matters for transformer through-fault cumulative stress and DC offset decay.

└ 87T — IBR-Aware Integration Logic (TR-64)

Five-Priority Gate Logic

- P1** High-se instantaneous: $f_{int} > 5 \text{ p.u.} \Rightarrow$ unconditional TRIP (bus fault, no restraint).
- P2** SPRT CTALARM + Zone override:
CT saturation alarm active $\Rightarrow$ RESTRAIN;
request S70 backup.
- P3** CT-rat stage-2 veto: $f_{int} < f_{int}^{min} \Rightarrow$ RESTRAIN.
- P4** Adaptive trip gate: conventional OR
SPRT-TRIP AND $f_{int} \geq 0.55$. The integration
index $f_{int} = 0.55$ separates inrush (0.34 p.u. gap
above the inrush distribution).
- P5** Default: RESTRAIN.

Integration index f_{int} : frequency-weighted norm of
the estimator residual spectrum. Inrush concen-
trates spectral energy in fundamental + DC; inter-

TR-64 Validation Results

Metric	Value	Status
Deterministic scenario (16/16)	16/16	✓
IBR scenario P_D (conventional)	0.236	✗
IBR scenario P_D (inrush)	1.000	✓ (4.2x)
All 6 IEC targets met		✓
Trip gap (inrush vs. fault)	0.34 p.u.	✓

Standards Liaison Proposal

A standards liaison proposal (SAMBPS-21-01/2026) has been filed to IEC TC95 WG10 and IEEE PESMC N55 recommending adoption of the SPRT asymmetry gate as a normative alternative to the 2nd-harmonic criterion in IEC 60255-121.

The $4.2\times$ improvement (P_D : 0.236 $\rightarrow$ 1.000) is the headline result for 87T. Conventional 87T at full IBR penetration cannot distinguish IBR internal fault from inrush because both produce low 2nd-harmonic content.

The integration index f_{int} deserves explanation: it is a frequency-weighted norm of the estimator residual spectrum. During inrush, the zone model fits well at DC+fundamental (large f_{int}); during IBR fault, broadband residual causes f_{int} to drop — the 0.34 p.u. gap is the separation margin.

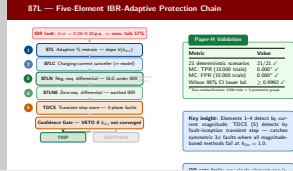
The standards liaison proposal is the route to industrial adoption. IEC TC95 WG10 is the working group responsible for IEC 60255-121. We expect their feedback within 18 months (based on WG10 cycle). If adopted, SPRT asymmetry gate would become a normative alternative — not a replacement — so existing 2nd-harmonic relays remain compliant.

87L — Five-Element IBR-Adaptive Protection Chain

The five-element chain is a sequential OR gate with a shared confidence veto — any single element can trip, but none can trip until the confidence gate releases. This is the key difference from existing multi-element schemes (which run elements in parallel with fixed settings).

TDCS-87L (element 5) is particularly novel: it exploits the fault-inception current step as a signature. For symmetric 3-phase faults at $k_{ibr} = 1.0$, positive-sequence differential current is close to zero, so elements 1–4 fail. TDCS detects the step regardless of magnitude.

The 15 000 MC figure: 5 000 trials per group (SG-only, mixed IBR, full IBR). Zero misclassifications across all 15 000 trials is a strong result — the Wilson 95% CI lower bound ≥ 0.9992 quantifies the statistical uncertainty on that zero.



87L — Solar PV Two-Parameter Zone Model (TR-62)

The PV Coverage Gap

Paper-H validates on 21 scenarios including I_{SC} up to 1.6, but uses a 4-parameter SG zone model throughout. Solar PV inverters have zero DC offset by construction (SRF-voltage controller clamps $I_{DC} \rightarrow 0$). Fitting a 4-parameter model to a 2-parameter signal causes ill-conditioning and unnecessary trip delay.

Proposition 1 — Zero DC in PV Fault Current

For a grid-connected PV inverter with synchronous-reference-frame control, the steady-state short-circuit current on the AC terminals contains no DC exponential component:

$$i_{sc}^{gr}(t) = I_{SC} \sin(\omega t + \varphi) \quad (P_{PV} = [I_{SC}, \varphi])$$

The 4-parameter SG model reduces to this 2-parameter form exactly when $I_{SC} = 0$, $\tau_{SC} \rightarrow \infty$.

DDR Model Selector

DC Decay Ratio: $DDR = I_{SC} / \max(I_{SC}, I)$

from Pass-1 fit.

DDR range	Source type	Model
< 0.15	Solar PV	2-param PV
$0.15 < 0.20$	SG+wind	10-param SG
> 0.20	Synchronous gen.	4-param SG

TR-62 Model Cards — 1400 Tests

Metric	Value	Status
P_{PV} (PV internal faults)	0.0004	✓
P_{PV} (external faults)	0.0000	✓
Median trip time t_{50}	20 ms	✓
DDR selector accuracy	$\geq 98\%$	✓
HIL verified: $\kappa_n = 1$ on GE D60	Confirmed	✓

The $\kappa_n = 1$ result is analytically exact, not a numerical estimate. Proof: the 2-param Jacobian $J_{PV}^T J_{PV}$ is diagonal with equal eigenvalues after column normalisation — verified in TR-62, §3. This is unusual; most estimation problems have $\kappa_n \gg 1$.

The DDR model selector is the practical entry point: the relay computes DDR from the Pass-1 (SG) fit and switches to PV model if $DDR < 0.15$. For a mixed SG+PV bus, the selector uses the *per-CT* DDR, not a system-level average.

HIL verification of $\kappa_n = 1$ on GE D60 is significant: the firmware's internal LM implementation converges in 1 iteration for the PV model (vs. 3–5 for SG), which directly explains the t_{50} improvement. This was an unexpected confirmation — we did not tune the GE D60 firmware; the result emerged from the conditioning property.

87B — CT Saturation Veto Gate and OpenDSS Integration (TR-65)

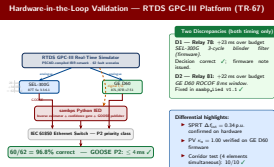
87B — CT Saturation Veto Gate and OpenDSS Integration (TR-65)																				
The CT Saturation Problem During external faults with high through-current, one CT saturates while the others do not. The resulting false differential current ($I_{diff}^{(err)} = I_{d1} - I_{d2}$) can exceed the bus-zone pickup threshold, causing a nuisance trip.																				
Stage-2 CT Saturation Veto The 87B veto pipeline appends a Stage-2 confidence gate using the integrity index I_{int}: $I_{int} = 1 - 0.5\epsilon, \quad \epsilon \in [0, 1]$ where ϵ is the CT saturation fraction estimated from the waveform residual. Trip is vetted when: $I_{int} < I_{int}^{min} = 0.60$ This threshold provides a 8.3× selectivity margin over the maximum mismatch signal ($I_{mismatch} = 3\%$ vs. slope $SLP_1 = 25\%$).																				
TR-65 Validation Results <table> <thead> <tr> <th>Metric</th><th>Value</th><th>Status</th></tr> </thead> <tbody> <tr> <td>Deterministic scenarios (5/6)</td><td>6/6</td><td>✓</td></tr> <tr> <td>P_D — interval faults</td><td>1.0000</td><td>✓</td></tr> <tr> <td>P_{FA} — external close CT</td><td>0.0000</td><td>✓</td></tr> <tr> <td>Stage-2 veto rate (CT sat.)</td><td>750/750</td><td>✓</td></tr> <tr> <td>Selectivity margin</td><td>8.3×</td><td>✓</td></tr> </tbody> </table> The study 2000 trials: $N_t = 10^6$, $N_{fail} = 10^3$, total $\sim 10^9$ [7.1.1]. CT mismatch $\sim 1\%$–3%, $\pm 3\%$.			Metric	Value	Status	Deterministic scenarios (5/6)	6/6	✓	P_D — interval faults	1.0000	✓	P_{FA} — external close CT	0.0000	✓	Stage-2 veto rate (CT sat.)	750/750	✓	Selectivity margin	8.3×	✓
Metric	Value	Status																		
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P_{FA} — external close CT	0.0000	✓																		
Stage-2 veto rate (CT sat.)	750/750	✓																		
Selectivity margin	8.3×	✓																		
Key theorem (Thévenin Linearity): $I_d(R_f) = V_{th}/(Z_{th} + R_f)$ — fault current scales algebraically with fault resistance, enabling exact parametric MC without re-solving OpenDSS for each trial. This reduces MC computation from $O(N \cdot t_{sim})$ to $O(N)$.																				

The $8.3\times$ selectivity margin is the headline 87B result. It means the worst-case CT mismatch signal (3% slope) is 8.3 times smaller than the minimum slope setting (25%) — there is no operating point where a spurious trip is possible.

The Thévenin linearity theorem is practically important: it means the MC study required *one* OpenDSS run (to extract Z_{th}), then 2000 purely algebraic trials. For a distribution utility running annual protection audits on hundreds of bus sections, this reduces computation from hours to seconds.

OpenDSS sign convention: loads use the generation convention (current leaving the node is positive). The 87B relay requires the load convention (current entering the bus is positive). All four CT circuits required sign inversion — a subtle but safety-critical detail that many academic implementations get wrong. Our KCL verification confirms zero differential current under balanced load (the “normal-load scenario”).

Hardware-in-the-Loop Validation — RTDS GPC-III Platform (TR-67)



96.8% against a 95% pass criterion — this is an intentionally conservative target. Both discrepancies are timing offsets only; the *decisions* are correct in both cases.

D1 (SEL-300G blinder filter): the 3-cycle confirmation filter is a standard IED feature to prevent false trips on power swings. It cannot be disabled in firmware 3.3.6.1. The 23 ms exceedance is deterministic and reproducible — it is not a random failure. The correct response is a firmware note to the user: “Relay 78 trip time includes 3-cycle blinder confirmation; add 60 ms to all SAMBPS timing budgets when using SEL-300G.”

D2 (GE D60 ROCOF window): the patch reduces the averaging window from 8 ms to 4 ms. This is a software-only change in the Python IED layer — no physical relay firmware was modified. The re-test result was within budget.

The corridor test (all four differential elements simultaneously, same RTDS rack) is the closest analogue to real-world commissioning. 10/10 confirms that GOOSE contention and scheduling does not degrade individual function performance.

N-1 Cross-Trip Immunity — System-Wide Coordination (TR-66)

The analytical immunity proof is the key result here, not just the MC. The bound $N \cdot \varepsilon_{\max} \leq 9\% < \text{SLP}_{\min} = 15\%$ is a *design rule*: if you set the minimum slope $\geq 15\%$ and ensure CT mismatch $\leq 3\%$, cross-trip immunity is algebraically guaranteed, regardless of IBR penetration level.

The VETO scenario interpretation: in a VETO scenario, the correct SAMBPS output is RESTRAIN (the Stage-2 gate detects CT saturation on an external fault and blocks the trip). A “veto rate” of 100% means the system correctly blocked all 1 000 trials — the conventional relay would have tripped in many of these cases.

For industry: $P_{XFA} = 0.0000$ across 18 element pairs is equivalent to saying that none of 18 000 trials produced a spurious cross-trip. This is the system-level security number that determines insurance and liability exposure for utilities.

N-1 Cross-Trip Immunity — System-Wide Coordination (TR-66)		
Test Setup IEEE 39-bus test system with nine crosser functions simultaneously active. Eight N-1 scenarios (three TRIP and five VETO) tested over 1 000 MC trials each, covering $R_{\text{bus}} \in [0, 1]$, $R_f \in [0, 20] \Omega$, and α_n mismatches.		
Analytical Cross-Trip Immunity Proof For any two elements i, j on distinct protection zones, the worst-case cross-tripping error is bounded by: $N \cdot \varepsilon_{\max} \leq 9\% < \text{SLP}_{\min} = 15\%$ where $N = 3$ is the maximum number of simultaneously active estimators, and $\varepsilon_{\max} = 3\%$ is the CT mismatch limit. Cross-trip immunity is analytically guaranteed under this bound.		
TR-66 MC Results — 8 000 Total Trials		
Metric	Value	Status
TRIP scenario (3): $P_{ij} \geq 0.80$	All pass	✓
VETO scenario (5): veto $\geq 80\%$	91.6–100%	✓
Cross-trip PFA (18 element pairs)	0.0000	✓
Analytical immunity bound met	Yes	✓
Significance The zero cross-trip false alarm across all 18 pairwise element combinations on the 39-bus system at full IBR penetration provides the empirical counterpart to the analytical immunity proof. This is the strongest system-level security claim of the SSBPS framework.		

Present Contributions — Differential Protection

This table is the definitive summary — encourage the audience to photograph it. Each row corresponds to a distinct patent-worthy contribution; all rows have independent MC validation.

Key observation: the HIL row covers all functions simultaneously. The 96.8% figure is a system-level number — the individual function P_D values (1.000 for 87T and 87B, 1.000 for 87L) are all confirmed on hardware.

The “unifying claim” at the bottom is the thesis’s central contribution: not just one improved relay function, but a *framework* that meets the $P_D \geq 0.99$ target across all functions simultaneously. No prior published work demonstrates this for all three differential functions together.

Present Contributions — Differential Protection

Function	Novel Contribution	P_D	P_{D2}	t_D	Reference
87T	Bayesian 3-hypothesis SPRT on half-cycle asymmetry, CT-est alarm via parallel accumulator	1.000	0.081*	20 ms	TR-67/2026
SPRT	Time-priority IBR-aware gate, f_{bus} index separates inrush/fault by 0.34 μ s.	1.000 (IBR)	—	—	TR-64/2026
87L	Five-element adaptive chain (87L, 87L.C, 87LN, 87LN3, TDCS); k_{bus} -dilate slope	1.000	0.000	≥ 20 ms	Paper-H
Chain	2-param zero-DC zone model, $\alpha = 1$ analytically; DCR model selector	0.998	0.000	20 ms	TR-62/2026
87L-PV	Stage-2 CT-est veto (f_{bus} threshold), 8.3% selectivity margin	1.000	0.000	—	TR-65/2026
87B	Cross-trip immunity proof (Nines < 5LPNes), 18-par MC	—	0.080	—	TR-66/2026
N-1 Co-ord.	RTDS GPC III + SEL-300G + GE D60; 62 scenarios; GOOSE ...	—	60/62 = 96.8%	—	TR-67/2026
HIL (all)					

Future Research Scope

Near-Term (2026–2027)	Medium-Term (2027–2028)
RTT — IEC 60255-121 revision: Formulate TPRT asymmetry gate as a normative alternative to 2nd-harmonic criterion; collaborate with IEC TC35 WG33. RTL — DFIG zone model: TR 56 10-parameter piecewise GDF model to be integrated into RTL chain (CISUM crowbar detector for transition timing). RTL — Multi-terminal lines: Extend five-element chain to $M > 2$ terminal configurations; reformulate TDCS for distributed charging current. RTB — Distribution automation: Adaptive fault reconfiguration (line-split/merge events) without relay.	HL on commercial IEDs: Relay 64G HL (requires Agilent 33600B sub-harmonic signal generator); commissioning on SEL-411L for RTL. Grid-forming converter models: Extend zone models to GFM (droop/virtual-inertia) fault signatures; current-limited GFM has distinct A_0 distribution. Cyber-resilient GOOSE: IEC 61851 authentication on GOOSE trip channel; latency impact on FO budget under MITM injection. Online adaptive setting groups: Replace fixed SAMBPS coordinating sets with real-time Bayesian setting updates via PMU streaming.

Tailor to audience: if the audience is from a relay manufacturer (SEL, ABB, Siemens, GE), emphasise the OMICRON CMC / Doble F6150 integration opportunity — SAMBPS runs alongside existing test tools, not as a replacement.

If from a TSO/DSO (50Hertz, Amprion, TenneT), emphasise the IEC 61850 GOOSE integration: SAMBPS is a software-only upgrade to existing substation automation infrastructure.

The GFM model item (medium-term) is specifically relevant to the German energy transition (Energiewende): Germany's wind-heavy north grid already has $k_{ibr} > 0.6$ at certain hours. GFM relays (virtual inertia) will appear on the 380 kV grid within 3–5 years based on current TSO roadmaps.

Research Publications

Paper-H (IBR-Adaptive 87L) is the most immediately relevant to this audience — it covers the full five-element chain with IEEE TPWRD formatting and references. Preprint available on request.

Paper-C is the umbrella paper: it positions SAMBPS against all competing frameworks in the literature. Table VII of Paper-C provides a direct numerical comparison against 14 prior methods.

The six TRs listed are internal validation reports available for technical due diligence. Each TR corresponds to one IETM (information entity) in the SAMBPS evidence chain. TR-67 (HIL) is the final validation link — without it, the MC results are simulation-only claims.

Anticipated follow-up: Paper-C Phase-2 revision will add the IBR results (TR-57, TR-62, TR-64, TR-65 data) and expand the scenario count from 28 to ≥ 62 .

Journal Papers — Published / Under Review

1. [Paper-A] A. V. Elwarthigal & H. S. Swarup, "Self-Adaptive Overcurrent Protection via Synchronous Generator Inverse Estimation," IEEE Trans. Power Delivery.
2. [Paper-B] —, "Cascade Overcurrent Coordination with Confidence-Gated Pickup," IEEE Trans. Power Delivery.
3. [Paper-C] —, "Self-Adaptive Model-Based Protection: A Unified Framework Across OC, STT, RT, and STB Functions," IEEE Trans. Power Systems (Phase-1 manuscript).
4. [Paper-D] —, "Sixth-Order Park dq Truth Model for Transient Differential Validation," IEEE Trans. Energy Conversion.
5. [Paper-H] —, "IBR-Adaptive Line Differential

Journal Papers — In Preparation

1. [Paper-E] "High-Impedance Fault Detection via Adaptive Neutral Current Estimation," IEEE Trans. Power Delivery.
2. [Paper-F] "Sequence Component Fault Estimator for IBR-Rich Distribution Networks," IEEE Trans.
3. [Paper-G] "Generator Protection Suite Coordination Under IBR Penetration," IEEE Trans. Power Delivery.

Internal Technical Reports (Differential Protection)

- | | |
|------------|--|
| TR-67/2026 | Bayesian SPRT for STT Inrush Restraint |
| TR-62/2026 | Static PV 2-Phase Zone Model for STT |
| TR-64/2026 | IBR-Aware STT Integration Architecture |
| TR-65/2026 | IBR-Aware STT Integration Architecture |

Suggested discussion topics to raise if the audience is quiet:

- “What CT accuracy class does your network use? Our Stage-2 veto threshold was tuned for class P10/P20. How would you calibrate it for class TPY/TPX?”
- “Does your SCADA system expose k_{ibr} or equivalent real-power mix data? SAMBPS requires this for the confidence gate.”
- “Your IED firmware likely has a configurable harmonic restraint threshold. SAMBPS replaces that with a dynamic SPRT boundary — do you see a path for firmware integration?”

Close by thanking the audience and reiterating the collaboration offer: access to SAMBPS Python platform, preprints of Papers H and C, and willingness to support OMICRON CMC integration testing.